PICT IEEE Student Branch – R&D Team

1. Objective

To cultivate a strong research-driven environment at PICT by enabling students to develop technical depth, contribute to impactful projects, and participate in global IEEE programs. The R&D team aims to empower first, second, and third-year students to collaboratively build solutions for real-world problems through hands-on research, sustainability initiatives, and industry-academic partnerships.

2. Quarter-wise Activity Plan (2026)

Q1: Foundations, Orientation & Collaborations (Jan–Mar)

A. Research Capacity Building

- **Research Orientation Bootcamp:** fundamentals of research, literature review, reproducibility, and technical writing.
- **Formation of Research Circles** in ML/AI, IoT & Embedded, Cybersecurity, Blockchain, Green Tech, and Systems.
- **Junior Integration:**
    - First-years join as learners and assistants (documentation, initial experimentation).
    - Second-years take responsibility for small modules or subcomponents.
    - Third-years act as research leads and project architects.

B. External Interfaces

- **Expert Lectures & Masterclasses** by college faculty, IIT/IISc, PICT Alumni researchers on AI, systems design, and sustainability engineering.
- **Roundtable on Research Careers** with alumni pursuing MS/PhD or working in R&D divisions.

C. Calls for Funding & Innovation

- Launch internal call for proposals for:

    - **IEEE New Initiatives Program (NIP)**
    - **IEEE EPICS (community-impact engineering)**
    - **Tech4Good Challenge**
- Teams formed across year groups; proposal writing sessions conducted.

Q2: Project Initiation, Skill Development & Sustainability (Apr–Jun)

A. Multi-Domain Research Projects Begin

Teams begin structured R&D projects, including:

1. **Smart Campus Sustainability System** – energy & water analytics, waste-tracking IoT, carbon footprint monitoring.
2. **AI-Assisted Healthcare Diagnostics** – medical imaging models, wearable health monitoring, signal analysis.

B. Skill-Building Workshops

- **Patent awareness & drafting workshop.**
- **IEEE Authorship Workshop** on writing technical papers.
- **Hands-on sessions** for Git, reproducibility, experiment tracking.

C. Sustainability Practices implementation

- **Green Labs Initiative:** reduced paper, optimized energy use; encourage low-power hardware.
- **E-Waste Repurposing for Projects:** old sensors, boards re-used in prototypes ask and seek PICT Robotics for consultations.
- **Solar Innovation Booth:** early design of a solar-powered kiosk for showcasing student R&D.

Q3: Research Review, Collaborations & Deployment (Jul–Sep)

A. PICT R&D Symposium (Internal Conference)

- Mid-project review through poster sessions, demos, and short papers.

- Judges from IITs/IISc and industry research groups invited.
- Selected teams continue to prepare for IEEE conference submissions.

B. Inter-Institute Collaborations

- **Research Mentorship Program**: quarterly guidance calls with PhD scholars from IITs/IISc.
- **Industry R&D Talks** by Microsoft Research India, TCS Research, Google Research India, Sarvam.ai.

Q4: Output, Funding Submissions (Oct–Dec)

A. Research Output & Documentation

- **Paper Writing Sprint** to prepare research manuscripts for IEEE conferences and student symposiums.
- **Patent Clinic (Advanced)** for teams with near-complete prototypes.
- Compilation of reproducible codebases and reports.

B. Grant Proposals & Funding

Teams finalize submissions to:

- IEEE NIP
- IEEE EPICS
- IEEE HAC/SIGHT
- Tech4Good
- IEEE Region 10 Grants

C. R&D Showcase & Sustainability Report

- **Annual Demo Day** with industry, alumni, and faculty to highlight all projects.
- Release of **PICT IEEE R&D Sustainability Annual Report** summarizing energy savings, green initiatives, and next year's goals.